

Seafloor Classification Guidelines for Students

Purpose

The goal of this exercise is to classify the seafloor into substrate classes using several raster layers and interpreted polygons.

Possible classes include:

Sand
Rock
Shallow rock
Mixed / transition zones

Classification should not be based on one image or one raster layer only. Each layer highlights a different physical or acoustic property of the seabed. A reliable interpretation comes from comparing all layers together.

Main Data Layers

The classification should be based on the following layers:

Bathymetry	= seafloor depth and morphology
Hillshade	= shaded relief / terrain texture
Slope	= local steepness and roughness
Backscatter	= acoustic reflectivity
Polygons	= manual interpretation / class boundaries

Together, these layers provide information about morphology, roughness, acoustic response, and interpreted substrate boundaries.

1. Bathymetry

Bathymetry represents the depth and general morphology of the seafloor.

What Is Clearly Visible

Large-scale structures
Ridges
Depressions
Elevated areas
General rocky morphology

What Is Less Visible

Subtle boundaries between sand and rock
Fine-scale texture
Flat rock surfaces
Small transition zones

Contribution to Classification

Bathymetry is useful for identifying the general shape of the seafloor. Irregular, elevated, or rough-looking areas may indicate rock or shallow rock. Smooth and uniform areas are more likely to represent sand.

However, bathymetry alone is not sufficient. Flat rock and sand may look similar if there is little vertical relief.

2. Hillshade

Hillshade enhances the three-dimensional appearance of the seafloor by simulating illumination.

What Is Clearly Visible

- Seafloor roughness
- Micro-relief
- Ridge edges
- Small scarps
- Rocky texture
- Shadow patterns

What Is Less Visible

- Acoustic reflectivity
- Material hardness
- Substrate type in very flat areas

Contribution to Classification

Hillshade is one of the most useful layers for identifying rock. Rocky areas usually show rough texture, small shadows, irregular edges, and clear micro-topography.

Sand usually appears smoother, with fewer shadows and weaker relief.

If an area appears rough in hillshade and also shows irregular morphology in bathymetry, it is likely to be rocky.

3. Slope

Slope shows how rapidly the seafloor elevation changes.

What Is Clearly Visible

- Sharp edges
- Local steep areas
- Rocky boundaries
- Rough terrain
- Small outcrops
- Transition zones

What Is Less Visible

Flat areas
Material type where relief is weak
Acoustic differences between substrates

Contribution to Classification

Slope is very useful for detecting rock edges and rough terrain. Rocky areas often produce higher and more variable slope values. Sand usually appears as smoother, low-slope terrain.

Slope should be used as an auxiliary layer. It helps confirm whether a visually rough area in hillshade or bathymetry also has real topographic variation.

4. Backscatter

Backscatter shows the strength of the acoustic signal returned from the seafloor.

What Is Clearly Visible

Differences between hard and soft substrate
Acoustic texture
High-reflectivity zones
Homogeneous versus heterogeneous areas

What Is Less Visible

Precise seafloor shape
Relative elevation
Detailed geomorphic boundaries

Contribution to Classification

Backscatter is critical for substrate interpretation. Hard substrates such as rock often produce stronger or more heterogeneous acoustic returns. Sand is usually more homogeneous and may show lower or smoother backscatter response.

However, backscatter should not be interpreted alone. It can be affected by survey geometry, incidence angle, noise, processing artifacts, and line direction.

The most reliable interpretation comes when backscatter patterns agree with bathymetry, hillshade, and slope.

5. Manual Polygons

Manual polygons represent interpreted class boundaries.

What They Represent

Sand boundaries
Rock boundaries
Shallow rock boundaries
Mixed or transitional zones

Contribution to Classification

The polygons are the interpreted labels. They can be used as reference data for manual classification or as training masks for supervised machine learning.

When drawing polygons, students should use all available layers. Boundaries should not be drawn from one visual layer alone.

How to Identify Each Class

Sand

Typical characteristics:

Bathymetry: smooth, flat, low relief

Hillshade: weak texture, few shadows

Slope: low and uniform values

Backscatter: relatively homogeneous acoustic texture

Sand is usually the most uniform class. It often forms broad, smooth areas with little terrain roughness.

Rock

Typical characteristics:

Bathymetry: irregular morphology, ridges, bumps, elevated areas

Hillshade: rough texture, shadows, sharp edges

Slope: higher and more variable slope values

Backscatter: stronger or more heterogeneous acoustic response

Rock is most confidently identified when several layers show the same pattern: rough morphology, high slope, clear hillshade texture, and distinctive backscatter.

Shallow Rock

Typical characteristics:

Bathymetry: weak to moderate relief

Hillshade: rough texture, but often less sharp than exposed rock

Slope: local slope changes, not always very strong

Backscatter: mixed or intermediate acoustic response

Shallow rock is more difficult to identify than exposed rock. It may appear as a transitional class between sand and rock.

Students should mark shallow rock only when the evidence is consistent across several layers.

Mixed or Transition Zones

Typical characteristics:

Unclear boundaries

Variable texture

Gradual changes between sand and rock
Conflicting evidence between layers

Mixed zones occur where the substrate is not clearly sand or rock. These areas should not be forced into a hard class if the evidence is uncertain.

Recommended Student Workflow

1. Start with bathymetry and identify the general seafloor morphology.
2. Look for large structures, ridges, depressions, and smooth areas.
3. Move to hillshade and examine roughness, shadows, and micro-relief.
4. Use slope to confirm sharp edges, local steepness, and rough terrain.
5. Use backscatter to check whether the acoustic response matches the morphological interpretation.
6. Draw polygons only where multiple layers support the interpretation.
7. Mark uncertain or gradual areas as transition zones.
8. Avoid using display colors alone. Interpret texture, shape, roughness, and acoustic response.

Important Rule

Do not classify an area using only one layer.

For example:

If backscatter is bright but hillshade is smooth and slope is low, the area may not be rock. It may be an acoustic effect.

If hillshade is rough, slope is high, and backscatter is distinct, the area is much more likely to be rock.

The strongest classification is obtained when several indicators agree:

Morphology
Roughness
Slope
Acoustic response
Manual interpretation

Summary Table

Layer	Main Contribution	Most Useful For	Main Limitation
Bathymetry	General morphology and depth	Large structures, ridges, relief	Weak for subtle substrate boundaries
Hillshade	Terrain texture and shadowing	Rock roughness, micro-relief	Does not measure acoustic response
Slope	Local steepness and edges	Rock boundaries, rough terrain	Less useful in flat areas
Backscatter	Acoustic reflectivity	Hard vs soft substrate contrast	Can be affected by survey artifacts
Polygons	Manual interpretation	Training labels and validation	Depends on interpreter quality

Final Notes

Reliable seafloor classification should be multi-layer and evidence-based.

In general:

Sand = smooth morphology + low slope + homogeneous texture

Rock = rough morphology + high slope + clear hillshade texture + distinct back scatter

Shallow rock = intermediate or partially covered rocky morphology

Transition zones = mixed or uncertain evidence

The best classifications are those where morphology, roughness, slope, and acoustic response support the same interpretation.